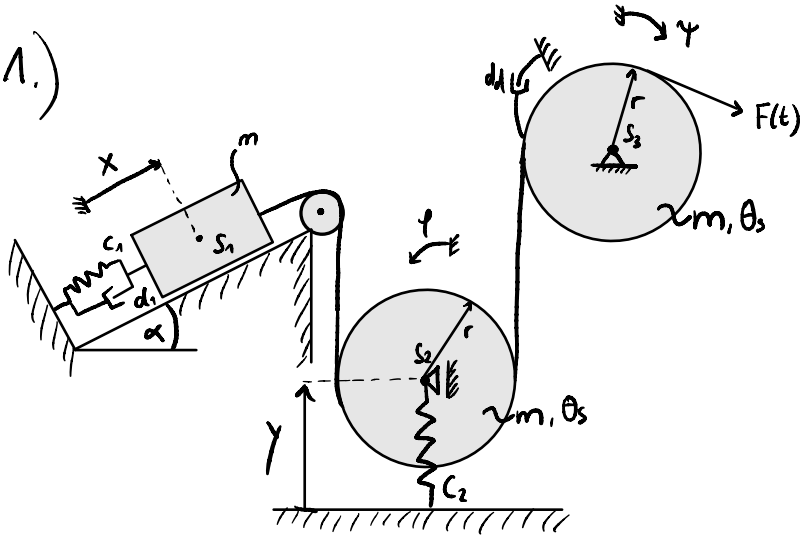
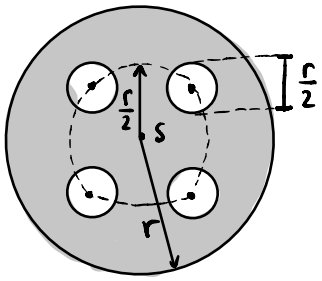
1.)




homogen




a) $\theta_s$ der Walzen bestimmen

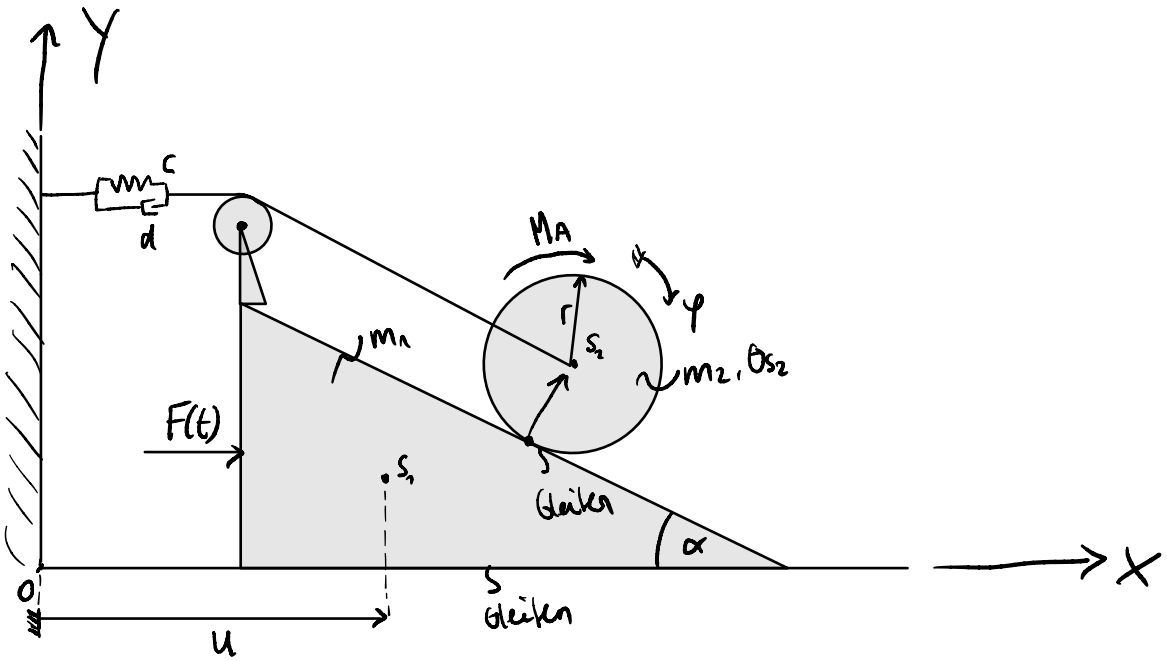
b) $\dot{\varphi}(\dot{x},\dot{\psi})$ $\dot{y}(\dot{x},\dot{\psi})$

c) FKB vom Allen Körpern
- Klotz Prinzip von d'lambert in x-Richtung
- Walze mit $S_2$ Drallsatz und P.v. d'lambert in y-Richtung
- Walze mit S3 Drallsatz

d) Schwingungen

e) Schwingung

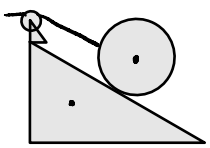
4)




a) $\underline{v}_{s_1}$, $\underline{v}_{s_2}$

b) $\delta x_{s_1}$, $\delta x_{s_2}$, $\delta y_{s_2}$

c)

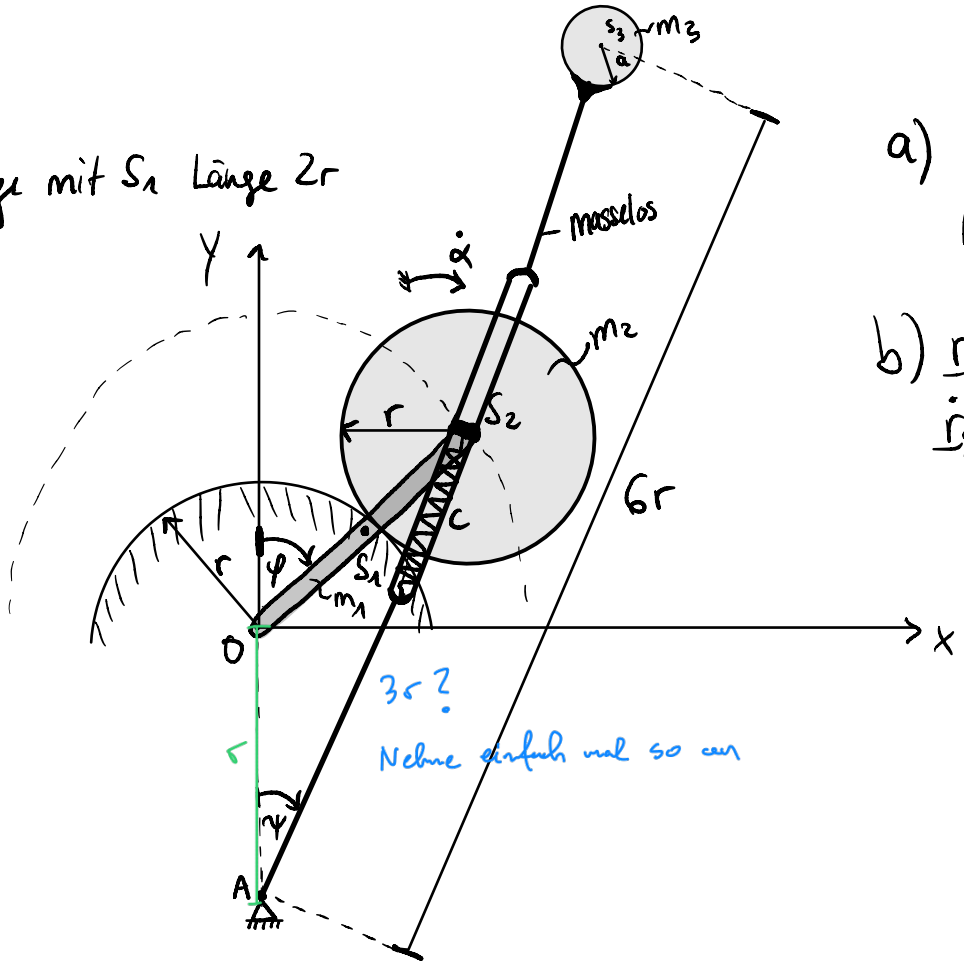


FKB mit allen benötigten Kräften für D'Alembert/Lagrange

d) Bewegungsgleichungen in u und φ

2)

- Stange mit $S_1$ Länge 2r




a) $\dot{\alpha}(\dot{\varphi})$
Abrollbedingung für ?

b) $\underline{r}_{S_1}, \underline{r}_{S_2}, \underline{r}_{S_3}$
$\underline{\dot{r}}_{S_1}, \underline{\dot{r}}_{S_2}, \underline{\dot{r}}_{S_3}$

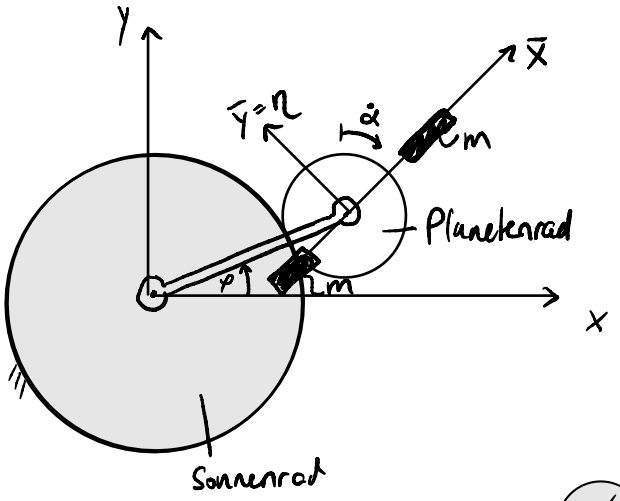
c) T und U aufstellen (Nullniveau X-Achse)

d) $\dot{\varphi}^2$ mit Energieerhaltungssatz

e) Berechne Federkonstante c, sodass es bei $\pi/2$ stehen bleibt.

3.)

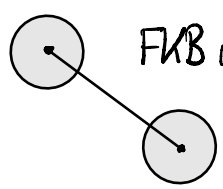
Draufsicht



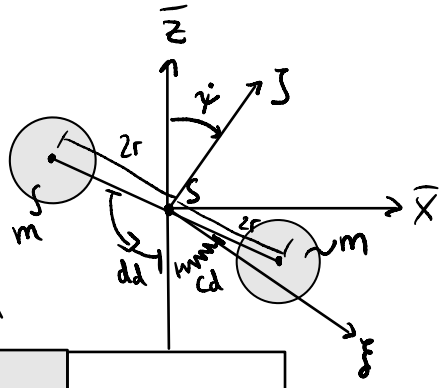

a) Kinematik

b) $e_\xi, e_\eta, e_\zeta$ in $(\bar{x},\bar{y},\bar{z})$ transformieren
$e_{\bar{x}} = ?$ $e_{\bar{y}} = ?$ $e_{\bar{z}} = ?$

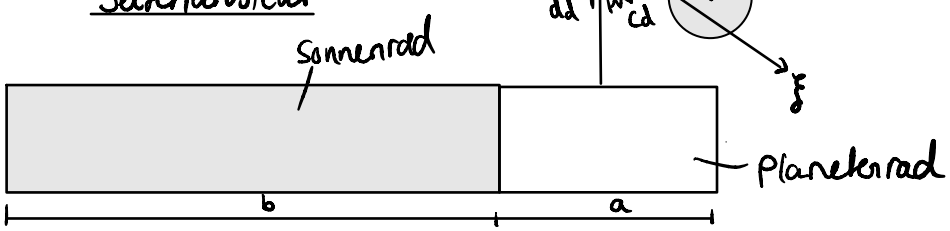
c) $\underline{\omega} = ?$ $\underline{\dot{\omega}} = ?$

d)  FKB und Drallsatz bzgl. S

e) Trägheitstensor




Seitenansicht



2)

a) $\underline{v}_{S_2} = \underline{v}_0 + \begin{pmatrix}0\\0\\-\dot\varphi\end{pmatrix} \times \begin{pmatrix}2r\sin(\varphi)\\2r\cos(\varphi)\\0\end{pmatrix} = \begin{pmatrix}2r\dot\varphi\cos(\varphi)\\-2r\dot\varphi\sin(\varphi)\\0\end{pmatrix}$

$|\underline{v}_{S_2}| = 2r\dot\varphi$

$\underline{v}_{S_2} = \underline{v}_M + \begin{pmatrix}0\\0\\-\dot\alpha\end{pmatrix} \times \begin{pmatrix}r\sin(\varphi)\\r\cos(\varphi)\\0\end{pmatrix} = \begin{pmatrix}r\dot\alpha\cos(\varphi)\\-r\dot\alpha\sin(\varphi)\\0\end{pmatrix}$

$|\underline{v}_{S_2}| = r\dot\alpha$

$|\underline{v}_{S_2}| \doteq |\underline{v}_{S_2}|$

$2r\dot\varphi = r\dot\alpha \Rightarrow \dot\alpha = 2\dot\varphi$

$v_{S_2} = \underline{v}_A + \begin{pmatrix}0\\0\\-\dot\psi\end{pmatrix} \times \begin{pmatrix}3r\sin(\psi)\\3r\cos(\psi)\\0\end{pmatrix}$

$v_{S_2} = 3r\dot\psi$

$3r\dot\psi = 2r\dot\varphi \Rightarrow \dot\psi = \frac{2}{3}\dot\varphi$

$\psi = \frac{2}{3}\varphi$

b)

$\underline{r}_{S_1} = \begin{pmatrix}r\sin(\varphi)\\r\cos(\varphi)\\0\end{pmatrix}$ $\underline{\dot r}_{S_1} = \begin{pmatrix}r\dot\varphi\cos(\varphi)\\-r\dot\varphi\sin(\varphi)\\0\end{pmatrix}$ $\dot r_{S_1}^2 = r^2\dot\varphi^2$

$\underline{r}_{S_2} = \begin{pmatrix}2r\sin(\varphi)\\2r\cos(\varphi)\\0\end{pmatrix}$ $\underline{\dot r}_{S_2} = \begin{pmatrix}2r\dot\varphi\cos(\varphi)\\-2r\dot\varphi\sin(\varphi)\\0\end{pmatrix}$ $\dot r_{S_2}^2 = 4r^2\dot\varphi^2$

$\dot r_{S_3}^2 = 16r^2\dot\varphi^2$

$\underline{r}_{S_3} = \begin{pmatrix}5r\sin(\psi)\\5r\cos(\psi)\\0\end{pmatrix} = \begin{pmatrix}5r\sin(\frac{2}{3}\varphi)\\5r\cos(\frac{2}{3}\varphi)\\0\end{pmatrix}$ $\underline{\dot r}_{S_3} = \begin{pmatrix}5r\dot\psi\cos(\psi)\\-5r\dot\psi\sin(\psi)\\0\end{pmatrix} = \begin{pmatrix}\frac{10}{3}r\dot\varphi\cos(\frac{2}{3}\varphi)\\-\frac{10}{3}r\dot\varphi\sin(\frac{2}{3}\varphi)\\0\end{pmatrix}$

c)

$T_{S_1} = \frac{1}{2}m_1\dot r_{S_1}^2 + \frac{1}{2}\Theta_{S_1}\dot\varphi^2 = \frac{1}{2}m_1r^2\dot\varphi^2 + \frac{1}{6}m_1r^2\dot\varphi^2 = \frac{2}{3}m_1r^2\dot\varphi^2$

$T_{S_2} = \frac{1}{2}m_2\dot r_{S_2}^2 + \frac{1}{2}\Theta_{S_2}\dot\alpha^2 = \frac{1}{2}m_2 4r^2\dot\varphi^2 + \frac{1}{4}m_2r^2 4\dot\varphi^2 = 3m_2r^2\dot\varphi^2$

$T_{S_3} = \frac{1}{2}m_3\dot r_{S_3}^2 + \frac{1}{2}\Theta_{S_3}\dot\gamma^2 = \frac{5}{3}m_3r^2\dot\varphi^2 + \frac{1}{5}m_3a^2\dot\varphi^2$

$T = \frac{2}{3}m_1r^2\dot\varphi^2 + 3m_2r^2\dot\varphi^2 + 8m_3r^2\dot\varphi^2 + \frac{1}{5}m_3a^2\dot\varphi^2$

$\dot r_{S_2}^2 = 4r^2\dot\varphi^2$ $\dot\psi = \frac{2}{3}\dot\varphi$ $\dot\alpha = 2\dot\varphi$

$U = \frac{1}{2}c\left(2r - 2r\cos(\psi)\right) + m_1g\cos(\varphi)\cdot r + m_2g\cdot\cos(\varphi)\cdot 2r + m_3g\left(6r\cos(\psi) - r\right)$ (?)

$U = \frac{1}{2}c\left(2r - 2r\cos(\frac{2}{3}\varphi)\right) + m_1gr\cos(\varphi) + m_2gr\cos(\varphi)\cdot 2r + m_3gr\left[6\cos(\frac{2}{3}\varphi) - 1\right]$

d)

$T_0 = 0$ Annahme $m_1 = m_2 = m_3 = m$

$U_0 = m_1gr + 2m_2gr + 5m_3gr = 8mgr$

$T_1 = \frac{35}{3}mr^2\dot\varphi^2 + \frac{1}{5}ma^2\dot\varphi^2 = \dot\varphi^2\left(\frac{35}{3}mr^2 + \frac{1}{5}ma^2\right)$

$U = \frac{1}{2}c\left(2r - 2r\cos(\frac{2}{3}\varphi)\right)^2 + 3mgr\cos(\varphi) + 6mgr\cos(\frac{2}{3}\varphi) - mgr$

$T_1 + U_1 = T_0 + U$

$\Rightarrow \dot\varphi^2\left(\frac{35}{3}mr^2 + \frac{1}{5}ma^2\right) = -\frac{1}{2}c\left(2r - 2r\cos(\frac{2}{3}\varphi)\right)^2 - 3mgr\cos(\varphi) - 6mgr\cos(\frac{2}{3}\varphi) + 8mgr - mgr$

$$\dot\varphi^2(\varphi) = \frac{-\frac{1}{2}c\left(2r - 2r\cos(\frac{2}{3}\varphi)\right) - 3mgr\cos(\varphi) - 6mgr\cos(\frac{2}{3}\varphi) + 8mgr - mgr}{\frac{35}{3}mr^2 + \frac{1}{5}ma^2}$$

$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

e) $\dot{\varphi}^2 \overset{!}{=} 0$ $\quad -\frac{1}{2}c\left(2r - 2r\cos\left(\frac{2}{3}\varphi\right)\right)^2 - 3mgr\cos(\varphi) - 6mgr\cos\left(\frac{2}{3}\varphi\right) + 8mgr - mgr = 0$

$$-\frac{1}{2}c\left(4r^2 + 4r^2\cos^2\left(\frac{2}{3}\varphi\right) - 8r^2\cos\left(\frac{2}{3}\varphi\right)\right) - 3mgr\cos(\varphi) - 6mgr\cos\left(\frac{2}{3}\varphi\right) + 7mgr = 0$$

$$c = \frac{-3mgr\cos(\varphi) - 6mgr\cos\left(\frac{2}{3}\varphi\right) + 7mgr}{2r^2 + 4r^2\cos^2\left(\frac{2}{3}\varphi\right) - 4r^2\cos\left(\frac{2}{3}\varphi\right)}$$

$$c\left(\varphi = \frac{\pi}{2}\right) = \frac{-3mgr\cos\left(\frac{\pi}{2}\right) - 6mgr\cos\left(\frac{\pi}{3}\right) + 7mgr}{2r^2 + 4r^2\cos^2\left(\frac{\pi}{3}\right) - 4r^2\cos\left(\frac{\pi}{3}\right)}$$

$$c\left(\varphi = \frac{\pi}{2}\right) = \frac{-3mgr + 7mgr}{2r^2 + r^2 - 2r^2} = \frac{4mgr}{r^2} = 4\frac{mg}{r}$$

Kp ob richtig :